

VARSHA SATHISKUMAR

Denver, CO | (720)-255-6441 | varsha_sathiskumar@mines.edu | [linkedin.com/in/varsha-sathiskumar](https://www.linkedin.com/in/varsha-sathiskumar) | github.com/VarshaSathiskumar

PROFESSIONAL SUMMARY

Software Engineer with 4+ years of experience building scalable full-stack enterprise and B2B applications, recently expanding into Agentic AI systems. Skilled at taking features from design through production and building reliable software that delivers measurable business impact.

EDUCATION

Colorado School of Mines

Master of Science in Computer Science - GPA: 3.6/4.0

| Golden, CO

Dec 2026

Thiagarajar College of Engineering

Bachelor of Technology in Information Technology - GPA: 8.7/10.0

| Madurai, India

Apr 2021

WORK EXPERIENCE

AI Software Engineer Intern | NociQuant Inc, Minnesota City, MN

Jun 2026 – Aug 2026

- Developed a full-stack healthcare application for reviewing patient findings, AI-generated differential diagnoses, and medical evidence, supporting earlier diagnosis targeting \$371.4B in potential chronic-pain savings using React and TypeScript.
- Engineered a RAG pipeline across 1,700+ medical papers covering 59 diseases, improving recall by 20% across 1,000 cases and increasing evidence completeness for diagnostic agents using LangChain, Pinecone, and Sentence-Transformers.
- Architected a multi-agent diagnostic debate system, improving diagnostic accuracy from 68% to 78% across 800 clinical cases spanning 11 diseases, using likelihood-ratio ranking, LangGraph, Gemini Lite/Pro and GPT models.
- Automated verification across 6 diagnostic configurations using 20+ metrics, including precision, recall, MRR, and ground-truth probability validation, identifying performance gaps through Python-based evaluation and scoring.

Full Stack Software Engineer | Kimberly Clark, India

May 2023 – Mar 2025

- Owned full-stack development of an enterprise pricing application, delivering 50+ features across 4 global regions and processing 200+ pricing agreements weekly, achieving over 97% user acceptance using C# and ASP.NET Core.
- Engineered 10+ REST APIs integrating SAP and Salesforce across 4 global regions, supporting 15K+ monthly pricing transactions while improving maintainability, security, and data access using Node.js and Redis.
- Optimized SQL Server performance across 4 databases and regions, reducing API response time from 5 to 4.4 minutes across 1000 test cases and reducing 10K record report downloads to 30 seconds.
- Modernized a legacy enterprise application into 4 independently deployed microservices, reducing synchronization failures and production support incidents by 32% using Docker and Kubernetes.

Associate Software Engineer | Kimberly Clark, India

Sep 2021 – Apr 2023

- Developed IoT monitoring dashboards supporting 5,000+ users across 33M+ sq. ft. in 15+ U.S. states and Canada, contributing to 105% higher visitor satisfaction and 80% lower consumable waste using Python, React and JavaScript.
- Built serverless Azure pipelines processing 1M+ daily IoT events across 2,000+ devices, cutting API response times from 40 to 20 seconds and eliminating 15 monthly complaints and work orders while saving \$68,052 annually per user.
- Implemented automated testing across 100+ workflows for 1 enterprise web and mobile application, expanding test cases from 1,200 to 1,800+ while identifying pre-release defects and regressions using Jest, pytest, and React Testing Library.

Software Engineer Intern | Kimberly Clark, India

Jan 2021 – Aug 2021

- Designed and deployed a 0→1 centralized consulting workflow serving 3,000+ global team members and processing 75 weekly requests, reducing processing from 1 day to 4 hours while standardizing request coordination using Power Platform Tool.

PROJECTS

Geospatial Web Platform

| Jan 2026 – May 2026

- Engineered a GIS platform reducing geological data query time from 40 mins to 2 secs and enabling interactive visualization across 12 map layers for Delaware Mountain Group exploration using React, FastAPI, and Mapbox GL JS.
- Developed ETL pipeline consolidating 2TB+ of geological data from 30+ source files and processing 10GB+ per execution, eliminating manual consolidation and standardizing geospatial data storage using PostgreSQL, Supabase, and PostGIS.

TECHNICAL SKILLS

Programming Languages & Frameworks: Python, C#, SQL, JavaScript, TypeScript, HTML, CSS, React, Node.js, FastAPI, Jest, pytest, Playwright, Selenium, Cypress

Development Tools & Platforms: PostgreSQL, Microsoft SQL Server, MySQL, Supabase, Azure Cosmos DB, Redis, AWS, Microsoft Azure, Docker, Kubernetes, Azure DevOps, Git, GitHub, Jira, Postman, Swagger/OpenAPI

AI/ML Tools: TensorFlow, Keras, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, Hugging Face, Sentence-Transformers, LangChain, LangGraph, MCP, Jupyter Notebook, Pinecone, Claude Code